

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA 02 COURSE NAME: Fundamentals Of Machine Learning

COURSE OUTCOME COVERED

CO5: *Create graphical models to analyze and infer patterns in complex probabilistic systems for structured decision-making.(BL5)*

Assessment Tool Weightage: 10%

QUESTIONS: 10

Total Marks:50

Annexure A

Mock Interview

Code of Conduct:

*I _S Sahil - 192525120_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Sahil

Annexure A

Mock Interview

- Q1. “Suppose you are working on an industrial monitoring system. How would you use Bayesian Networks to predict machine failures?” (5 Marks – 8 Mins)
- Q2. “Imagine you are asked in an interview to design an image denoising solution. How would Markov Random Fields help model neighborhood pixel dependencies?” (5 Marks – 8 Mins)
- Q3. “If a company wants to build a speech recognition system, how would you apply a Hidden Markov Model to process sequential audio signals?” (5 Marks – 8 Mins)
- Q4. “During a technical interview, how would you explain the use of Conditional Random Fields for sequence labeling tasks such as part-of-speech tagging?” (5 Marks – 8 Mins)
- Q5. “Suppose an interviewer asks how conditional independence simplifies probabilistic inference. How would you justify it with a graphical model example?” (5 Marks – 8 Mins)
- Q6. “In a healthcare diagnosis problem, how would you compare directed graphical models and undirected graphical models for representing symptom relationships?” (5 Marks – 8 Mins)
- Q7. “If patient symptom data is partially missing, how would you perform Bayesian inference to still make an accurate diagnosis?” (5 Marks – 8 Mins)
- Q8. “How would you use a Hidden Markov Model to analyze customer purchase sequences in an e-commerce recommendation system?” (5 Marks – 8 Mins)
- Q9. “In an NLP interview task, how would you explain why Conditional Random Fields often outperform Hidden Markov Models in named entity recognition?” (5 Marks – 8 Mins)
- Q10. “Suppose you are asked to improve fraud detection. How would graphical model learning and generalization enhance prediction accuracy?” (5 Marks – 8 Mins)

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professionalism	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

		behavior throughout			
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Q1. “Suppose you are working on an industrial monitoring system. How would you use Bayesian Networks to predict machine failures?” (5 Marks – 8 Mins)

Answer:

A Bayesian Network is a probabilistic graphical model that represents the relationship between different variables. It is useful for predicting machine failures because an industrial machine depends on many factors such as temperature, vibration, pressure, oil level, machine load, and operating time. These factors can be represented as nodes in the Bayesian Network, and the relationships between them are represented using directed arrows.

First, we collect historical data from the machines. This data contains sensor readings and information about whether the machine worked normally or experienced a failure. From this data, the Bayesian Network learns the probability relationships between different machine conditions. For example, it may learn that when temperature is very high and vibration is also high, the probability of machine failure increases.

During actual operation, sensors continuously provide new information about the machine. These readings are given to the Bayesian Network as evidence.

The network then uses Bayesian inference to calculate the probability of failure based on the available sensor information.

For example, if the temperature, vibration, and pressure values become abnormal, the probability of failure may become high.

If the calculated probability crosses a predefined threshold, the system can send an early warning to the operator. Maintenance can then be performed before the machine completely breaks down. This helps prevent unexpected failures and reduces production downtime.

Therefore, Bayesian Networks can be used for early failure prediction, preventive maintenance, risk analysis, and improving machine reliability. The main advantage is that the model can combine different uncertain sensor readings and provide a probability-based prediction instead of giving only a simple yes-or-no result.

Q3. “If a company wants to build a speech recognition system, how would you apply a Hidden Markov Model to process sequential audio signals?” (5 Marks – 8 Mins)

Answer:

A Hidden Markov Model (HMM) is a probabilistic model that is very useful for handling sequential data. Speech recognition is a good example because speech is not a single signal; it is a sequence of sounds that changes continuously with time. Therefore, an HMM can be used to understand the sequence of sounds and convert them into words.

First, the speech signal is collected through a microphone and divided into many small time frames. From each frame, useful audio features are extracted. These features may represent characteristics of the speech signal and are treated as the observations in the HMM.

The actual speech sounds cannot be directly identified from the observations, so they are represented as hidden states. These states can represent phonemes or other speech units. The HMM learns the probability of moving from one state to another. For example, it learns which sound is likely to come after another sound. It also learns the probability of an observed audio feature being produced by a particular hidden speech state.

When a person speaks a sentence, the HMM receives a sequence of audio observations.

It compares these observations with the probabilities learned during training and tries to find the most probable sequence of hidden speech states. The Viterbi algorithm can be used to find this most likely sequence.

After identifying the sequence of speech sounds, the system maps them to words and produces the final recognized text. For example, the audio signal from a person saying “machine learning” is processed as a sequence of observations, and the HMM determines the most probable sequence of speech units corresponding to those words.

Therefore, HMMs are useful in speech recognition because they can model time-dependent sequences, hidden speech states, and transitions between sounds. They allow the system to process speech step by step and determine the most probable words from the audio signal.

Q9. “In an NLP interview task, how would you explain why Conditional Random Fields often outperform Hidden Markov Models in named entity recognition?” (5 Marks – 8 Mins)

Answer:

Conditional Random Fields (CRFs) are probabilistic models commonly used for sequence labeling tasks, including Named Entity Recognition (NER). NER is used in Natural Language Processing to identify important entities such as people, organizations, locations, companies, and dates from a sentence. For example, in the sentence “Sahil joined Google in India,” the system needs to identify “Sahil” as a person, “Google” as an organization, and “India” as a location.

CRFs often perform better than basic Hidden Markov Models (HMMs) because they can use many different features from the input text. For example, a CRF can consider the current word, previous word, next word, capitalization, prefixes, suffixes, punctuation, and surrounding context. These features provide more information to the model when deciding the correct label for a word.

Another important advantage is that CRFs consider the relationship between neighboring output labels. In NER, labels are not completely independent. For example, if one word is identified as the beginning of a person's name, the following word may also belong to the same person's name. CRFs can use these relationships to produce a more consistent sequence of labels.

HMMs, on the other hand, are generative models that model both the hidden states and the observed words. Basic HMMs make stronger independence assumptions about the observations. Because of these assumptions, it can be difficult for an HMM to effectively use many different and overlapping text features.

CRFs directly model the probability of the label sequence given the observed text. This allows them to focus on choosing the best labels for the given sentence. As a result, CRFs can make better use of contextual information and complex features.

Therefore, CRFs often outperform HMMs in Named Entity Recognition because they can use more features, more context, and dependencies between neighboring labels.